

# JAEOON JUNG

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## Summary

I received my B.S. in Artificial Intelligence from Soongsil University and work at MAUM.AI as a senior AI research scientist. My research focuses on building intelligent agents that perceive, reason, and act across diverse modalities, with interests in embodied AI, automated fact-checking, and multimodal understanding.

## Education

### Soongsil University

Mar 2020 – Feb 2026

*Bachelor of Science in Artificial Intelligence*

*Seoul, South Korea*

- Includes a 2-year mandatory alternative military service at MAUM.AI: Apr 2023 – Mar 2025
- GPA 4.2 / 4.5, Summa Cum Laude

## Research Experience

### MAUM.AI

Dec 2024 – Current

*Senior AI Research Scientist*

*Pangyo, South Korea*

- Working on [WoRV team](#) to build a End-to-End Vision-Language Action model to drive or act like human. [C5][C6][P1][P3] Collaborating with SNUPI Lab, led by Prof. Youngjae Yu.
- Built surveillance VLMs for real-time video anomaly detection on edge devices such as Jetson NX and Hailo-10H.

*AI Research Scientist*

Apr 2023 – Nov 2024

- Contributed to vision-language action models in team WoRV, enabling commonsense reasoning-based VLA navigation using only visual perception [C2, W3], and deployed in agricultural robots.
- Led the development of a Korean VLM and a comprehensive benchmark framework for evaluating Korean VLMs. [W4]
- Led continual pre-training to transfer instruction-following capabilities from English to Korean using open-source LLMs, enabling low-cost development of high-quality Korean language models and deployed in 5+ commercial services, including the Korea Ministry of Employment and Labor's AI Labor Law Consultation.
- Experienced with training large language models(up to 70B), using deepspeed for multi-node training. While each node comprises 8 NVIDIA H100 80GB GPUs, eight DGX H100 systems interconnected with NVSwitch are used for training.

### HUMANE Lab

Jun 2024 – Feb 2026

*Undergraduate research student advised by Kunwoo Park*

*Soongsil University, South Korea*

- Worked on automated fact-checking agent for both text-only [W2][W5] and multimodal settings [C7, W6][W7].

### Replica AI

Mar 2025 – Apr 2025

*ML Engineer, independent contractor*

*Remote, United State*

- Worked on virtual try-on (VITON) solutions for fashion tech.

### MISys Lab

Jun 2021 – Aug 2022

*Undergraduate research student advised by Kanghee Kim*

*Soongsil University, South Korea*

- Studied C++, developed EtherCAT-based motor control systems, and built an electric go-kart.

## Publications

(\*: Equal contribution, C: Conferences, J: Journal, W: Workshop, R: Competition Report, P: Preprint)

### Preprints

[P3] **PonderPounce: A Pretrained MLLM as an Episode Context Engine for Robot Control**

Suhwan Choi\*, **Jaeyoon Jung\***, Sungkyung Kim, Yunsung Lee, Youngjae Yu

*Under Review*

[P2][R2] **The PokeAgent Challenge: Competitive and Long-Context Learning at Scale**

Seth Karten, Jake Grigsby, Tersoo Upaa Jr, Junik Bae, Seonghun Hong, Hyunyoung Jeong, **Jaeyoon Jung**, Kun Kerdtthaisong, Gyungbo Kim, Hyeokgi Kim, Yujin Kim, Eunju Kwon, Dongyu Liu, Patrick Mariglia, Sangyeon Park, Benedikt Schink, Xianwei Shi, Anthony Sistilli, Joseph Twin, Arian Urdu, Matin Urdu, Qiao Wang, Ling Wu, Wenli Zhang, Kunsheng Zhou, Stephanie Milani, Kiran Vodrahalli, Amy Zhang, Fei Fang, Yuke Zhu, Chi Jin

*Under Review*

*NeurIPS 2025 Competition Report*

- [P1][W9] **WorldCam: Interactive Autoregressive 3D Gaming Worlds with Camera Pose as a Unifying Geometric Representation**  
Jisu Nam, Yicong Hong, Chun-Hao Paul Huang, Feng Liu, JoungBin Lee, Jiyoung Kim, Siyoon Jin, Yunsung Lee, **Jaeyoon Jung**, Suhwan Choi, Seungryong Kim, Yang Zhou  
*Under Review*  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2026*

#### Conferences

- [C7][W6] **Is a Picture Worth a Thousand Words? Adaptive Multimodal Fact-Checking with Visual Evidence Necessity**  
**Jaeyoon Jung**, Yejun Yoon and Kunwoo Park  
*Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (ACL-IJCNLP Main), 2026*  
*Workshop on Human-Centric AI at CIKM, 2025*
- [C6][W8] **CostNav: A Navigation Benchmark for Cost-Aware Evaluation of Embodied Agents**  
Haebin Seong, Sungmin Kim, Minchan Kim, Yongjun Cho, Myunchul Joe, Suhwan Choi, **Jaeyoon Jung**, Jiyong Youn, Yoonshik Kim, Samwoo Seong, Yubeen Park, Youngjae Yu, Yunsung Lee  
*Conference on Robot Learning (CoRL), 2026*  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2026*
- [C5] **D2E: Scaling Vision-Action Pretraining on Desktop Data for Transfer to Embodied AI**  
Suhwan Choi\*, **Jaeyoon Jung\***, Haebin Seong\*, Minchan Kim, Minyeong Kim, Yongjun Cho, Yoonshik Kim, Park Yu Been, Youngjae Yu, and Yunsung Lee  
*International Conference on Learning Representations (ICLR), 2026*
- [C4] **Exploring Fine-Tuning of Large Audio Language Models for Spoken Language Understanding under Limited Speech Data**  
Youngwon Choi, **Jaeyoon Jung**, Hyeonyu Kim, Huu-Kim Nguyen, Hwayeon Kim  
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (Oral), 2026*
- [C3] **Hypothetical Documents or Knowledge Leakage? Rethinking LLM-based Query Expansion**  
Yejun Yoon, **Jaeyoon Jung**, Seunghyun Yoon, Kunwoo Park  
*Findings of the Association for Computational Linguistics (ACL Findings), 2025*
- [C2][W3] **CANVAS: Commonsense-Aware Navigation System for Intuitive Human-Robot Interaction**  
Suhwan Choi\*, Yongjun Cho\*, Minchan Kim\*, **Jaeyoon Jung\***, Myunchul Joe, Yubeen Park, Minseo Kim, Sungwoong Kim, Sungjae Lee, Hwiseong Park, Jiwan Chung, Youngjae Yu  
*IEEE International Conference on Robotics and Automation (ICRA), 2025*  
*Workshop on Open-World Agents at NeurIPS (Oral, Outstanding Paper Awards), 2024*
- [C1] **EnCLAP: Combining Neural Audio Codec and Audio-Text Joint Embedding for Automated Audio Captioning**  
Jaeyeon Kim, **Jaeyoon Jung**, Jinjoo Lee, Sang Hoon Woo  
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024*

#### Workshops

- [W7] **VILLAIN at AVerImaTeC: Verifying Image-Text Claims via Multi-Agent Collaboration**  
**Jaeyoon Jung\***, Yejun Yoon\*, Seunghyun Yoon, Kunwoo Park  
*Ninth Workshop on Fact Extraction and VERification (FEVER) at EACL (Oral), 2026*
- [W5] **Team HUMANE at AVeriTeC 2025: HerO 2 for Efficient Fact Verification**  
Yejun Yoon, **Jaeyoon Jung**, Seunghyun Yoon and Kunwoo Park  
*Eighth Workshop on Fact Extraction and VERification (FEVER) at ACL, 2025*
- [W4] **KOFFVQA: An Objectively Evaluated Free-form VQA Benchmark for Large Vision-Language Models in the Korean Language**  
Yoonshik Kim\* and **Jaeyoon Jung\***  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2025*
- [W2] **The Herd of Open LLMs for Verifying Real-World Claims**  
Yejun Yoon\*, **Jaeyoon Jung\***, Seunghyun Yoon, Kunwoo Park  
*Seventh Workshop on Fact Extraction and VERification (FEVER) at EMNLP (Oral), 2024*
- [W1] **EnCLAP++: Analyzing the EnCLAP Framework for Optimizing Automated Audio Captioning Performance**  
Jaeyeon Kim, Minjeon Jeon, **Jaeyoon Jung**, Sang Hoon Woo, Jinjoo Lee  
*Detection and Classification of Acoustic Scenes and Events (DCASE) Workshop, 2024*

## Competition Reports

[R1]      **Expanding on EnCLAP with Auxiliary Retrieval Model for Automated Audio Captioning**  
Jaeyeon Kim, **Jaeyoon Jung**, Minjeong Jeon, Sang Hoon Woo, Jinjoo Lee  
*DCASE 2024 Challenge Technical Report*

## Challenge

|                                                                                                                                                                                                                                                                                                                                                                                                              |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>1st Place</b> , 9th FEVER AVerImaTeC shared task (EACL 2026) <ul style="list-style-type: none"><li>Built multimodal fact-checking pipeline with agentic multimodal models</li></ul>                                                                                                                                                                                                                       | <b>Dec 2025</b> |
| <b>Judge's Choice Award</b> , PokéAgent Challenge at NeurIPS 2025 <ul style="list-style-type: none"><li>Award: \$400, competition report co-authorship</li><li>Participated in Track 2: speedrunning, long-horizon RPG gameplay</li></ul>                                                                                                                                                                    | <b>Nov 2025</b> |
| <b>1st Place</b> , 5th Uni-DTHON Challenge <ul style="list-style-type: none"><li>Award: ₩2,000,000 (<math>\approx</math> \$1400)</li></ul>                                                                                                                                                                                                                                                                   | <b>Nov 2025</b> |
| <b>2nd Place</b> , 8th FEVER AVeriTeC shared task (ACL 2025) <ul style="list-style-type: none"><li>We enhance our previous year's fact-checking pipeline using new retrieval method and quantization</li></ul>                                                                                                                                                                                               | <b>May 2025</b> |
| <b>2nd Place</b> , 7th FEVER AVeriTeC shared task (EMNLP 2024) <ul style="list-style-type: none"><li>We build our fact-checking pipeline with open LLMs, specifically llama in AVeriTeC fact-checking dataset.</li><li>Since our pipeline was developed with open LLMs, our system was selected as a baseline for the eight workshop.</li></ul>                                                              | <b>Aug 2024</b> |
| <b>2nd Place</b> , Detection and Classification of Acoustic Scenes and Events 2024 Challenge <ul style="list-style-type: none"><li>Participated in Task 6: Automated Audio Captioning, and Task 8: Language-Based Audio Retrieval.</li></ul>                                                                                                                                                                 | <b>Jun 2024</b> |
| <b>1st Place</b> , LG Display Product Quality Classification (LG AI Research) <ul style="list-style-type: none"><li>Award: ₩5,000,000 (<math>\approx</math> \$3500)</li><li>Built a robust regressor and classifier, then combined them with custom hard voting into a generalizable ensemble model</li></ul>                                                                                                | <b>Mar 2023</b> |
| <b>1st Place</b> , 2022 Samsung AI Challenge (3D Metrology) <ul style="list-style-type: none"><li>Award: ₩10,000,000 (<math>\approx</math> \$7000)</li><li>Used CycleGAN for unpaired transfer from simulated to real SEM images since sim and real SEM images were not paired, and improved depth map prediction using cosine similarity-based KNN to retrieve relevant depth maps as references.</li></ul> | <b>Oct 2022</b> |
| <b>2nd Place</b> , LG Innotek Radar Performance Prediction (LG AI Research) <ul style="list-style-type: none"><li>Award: ₩3,000,000 (<math>\approx</math> \$2100)</li><li>Analyzed radar factory tabular data using domain knowledge and SHAP to engineer and select features, then trained a boosting ensemble on the refined imbalanced dataset with stratified k-fold.</li></ul>                          | <b>Sep 2022</b> |
| <b>4th Prize</b> , 2023 SWUniv AI Challenge                                                                                                                                                                                                                                                                                                                                                                  | <b>Nov 2023</b> |
| <b>2nd Place</b> , 2022 Soongsil University AI Contest                                                                                                                                                                                                                                                                                                                                                       | <b>Nov 2022</b> |
| <b>7th Place</b> , 2022 SWUniv AI Challenge                                                                                                                                                                                                                                                                                                                                                                  | <b>Nov 2022</b> |
| <b>Reach the Final</b> , 2022 Naver AI RUSH                                                                                                                                                                                                                                                                                                                                                                  | <b>Aug 2022</b> |
| <b>7th Place</b> , 2022 Dankook University AI Challenge                                                                                                                                                                                                                                                                                                                                                      | <b>Jul 2022</b> |
| <b>Participation Prize</b> , 2021 Soongsil University AI Contest                                                                                                                                                                                                                                                                                                                                             | <b>Nov 2021</b> |

## Awards and Honors

|                                                                                                                                                                                                                                                                   |                    |
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| <b>Summa Cum Laude</b> , Department of Artificial Intelligence, Soongsil University                                                                                                                                                                               | <b>Feb 2026</b>    |
| <b>Best Thesis Award</b> , Department of Artificial Intelligence, Soongsil University <ul style="list-style-type: none"><li>With paper "Is a Picture Worth a Thousand Words? Adaptive Multimodal Fact-Checking with Visual Evidence Necessity" [C7, W6]</li></ul> | <b>Dec 2025</b>    |
| <b>Outstanding Paper Award</b> , NeurIPS 2024 - Open World Agent Workshop <ul style="list-style-type: none"><li>With paper "CANVAS: Commonsense-Aware Navigation System for Intuitive Human-Robot Interaction" [C2, W3]</li></ul>                                 | <b>Dec 2024</b>    |
| <b>Soongsil University Academic Scholarship</b> <ul style="list-style-type: none"><li>Scholarship of ₩6,000,000 (<math>\approx</math> \$4200)</li></ul>                                                                                                           | <b>2020 – 2022</b> |

## Activities

|                                                                                                                                           |                            |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Deepest: SNU Deep Learning Society</b> <ul style="list-style-type: none"><li>Season 19 Board Member</li><li>Season 18 Member</li></ul> | <b>Sep 2025 – Current</b>  |
| <b>Pseudo Lab: Season 11 Member</b>                                                                                                       | <b>Mar 2025 - Jun 2025</b> |

## Academic Services

Journal reviewer: IEEE Signal Processing Letters (SPL)